

Food poisoning

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Medical Microbiology and Immunology

Food poisoning

- An illness caused by eating contaminated food. It's not usually serious and most people get better within a few days without treatment. Except caused by Clostridium botulinum
- Can be spread by an infected food handler.



Food Poisoning



Abdominal pain

Diarrhea

Fever

Nausea
Vomiting

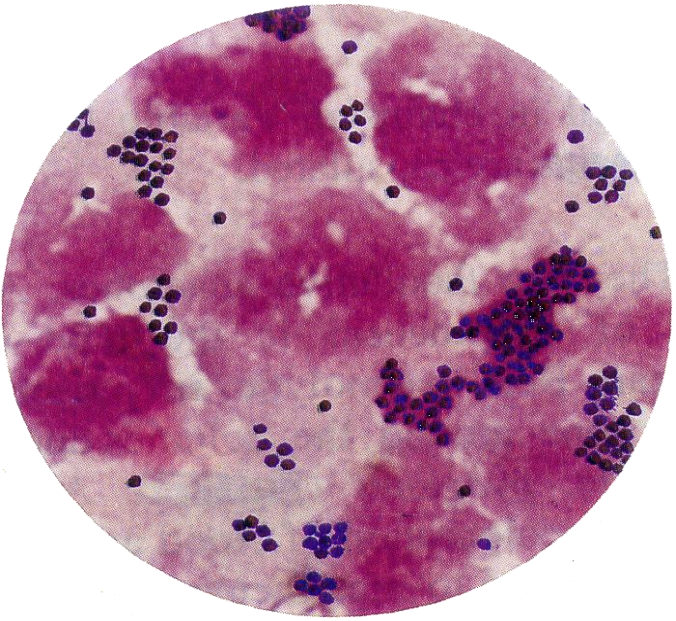
Malaise

Staphylococcus aureus

- **Incubation period:** 1 to 6 hours
- CHO, milk, milk products. Prepared salads, cream sauces, and cream-filled pastries.
- Can be spread by hand contact, coughing and sneezing.
- **Pathogenesis:** preformed enterotoxin in food

Identification of *Staphylococcus aureus:*

Direct microscopic examination: Gram-positive cocci arranged in clusters with numerous polymorph nuclear cells (PMNs)

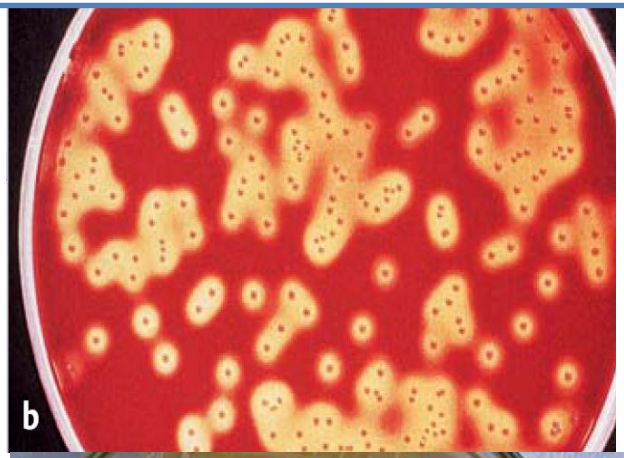


Culture (Incubated at 37 °C for 24 h).

β – hemolysis on blood agar

Golden yellow endopigment

On Nutrient agar:

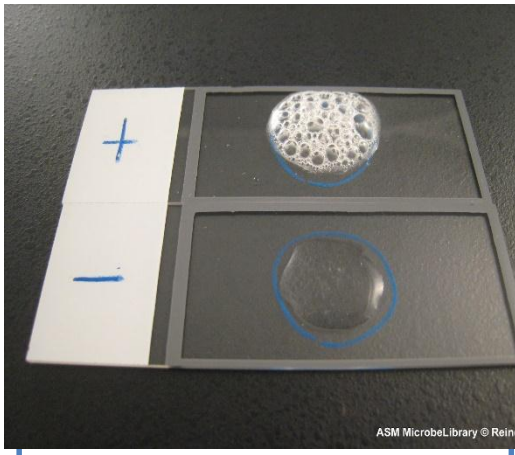




Biochemical reaction:

Coagulase positive





Biochemical reaction:

Catalse test: positive

How to trace the source of staphylococcal infections in case of food poisoning & How to detect enterotoxin ?

Increase your knowledge



Bacillus cereus (Emetic type)

- **I. P.:** 1 to 5 hours
- Can be spread through contaminated reheated cooked rice
- Pathogenesis: heat stable enterotoxin→vomiting

Bacillus cereus (Diarrheal type)

- **I. P.:** 8 to 24 hours
- Can be spread through contaminated meat
- Pathogenesis: heat labile enterotoxin→Diarrhea

Campylobacter

- Incubation period: 2 to 5 days
- Meat and undercooked poultry or you eat food that has touched raw or undercooked poultry. Other sources include unpasteurized milk and contaminated water
- **Pathogenesis:** Multiplication in gut that results in inflammation with red and white blood cells in stools.

Clostridium botulinum

- **Incubation period:** 12 to 72 hours
- No GIT manifestations but neurological manifestations
- Home-canned foods with low acidity, improperly canned commercial foods, smoked or salted fish. **Pathogenesis**: Neurotoxin in food
- **Prognosis:** high mortality rate

Clostridium perfringens

- **Incubation period:** 8 to 16 hours
- Precooked meat, contaminated by Clostridium perfringens spores
- **Pathogenesis:** Multiplication in gut & Enterotoxin

Escherichia coli (E. coli) O157:H7

- **Incubation period:** 1 to 8 days
- Spread mainly by undercooked ground beef. Other sources include unpasteurized milk and and contaminated water.
- **Pathogenesis:** Intimate adherence with formation of attaching and effacing lesions in gut & Shiga like toxin

Listeria monocytogenes

- **Incubation period:** 9 to 48 hours
- Hot dogs, luncheon meats, unpasteurized milk and cheeses.
- **Pathogenesis:** Multiplication in gut & Invasion and Listeriolysin O.

Salmonella

- This group features Salmonella enterica serotype Enteritidis and Salmonella enterica serotype Typhimurium
- **Incubation period:** 1 to 3 days
- Raw or contaminated meat, poultry, milk, or egg yolks. Survives inadequate cooking. Can be spread by knives, cutting surfaces or an infected food handler.
- **Pathogenesis:** Multiplication in intestine and colon

Shigella

- **Incubation period:** 24 to 48 hours
- Seafood and raw, ready-to-eat produce. Can be spread by an infected food handler.

Vibrio parahaemolyticus

- **Incubation period:** 1 to 7 days
- Can be spread through contaminated seafood or seawater.
- Pathogenesis: Multiplication in gut & toxin (hemolysin)

Dr Mohamed Sakr

Viruses causing food poisoning

- Ingestion of contaminated food and water.
- Hepatitis A **I. P.:** 28 days
- Noroviruses (like Norwalk viruses) (raw oysters) **I. P.:** 12 to 48 hours
- Rotavirus **I. P.:** 1 to 3 days


Thank you